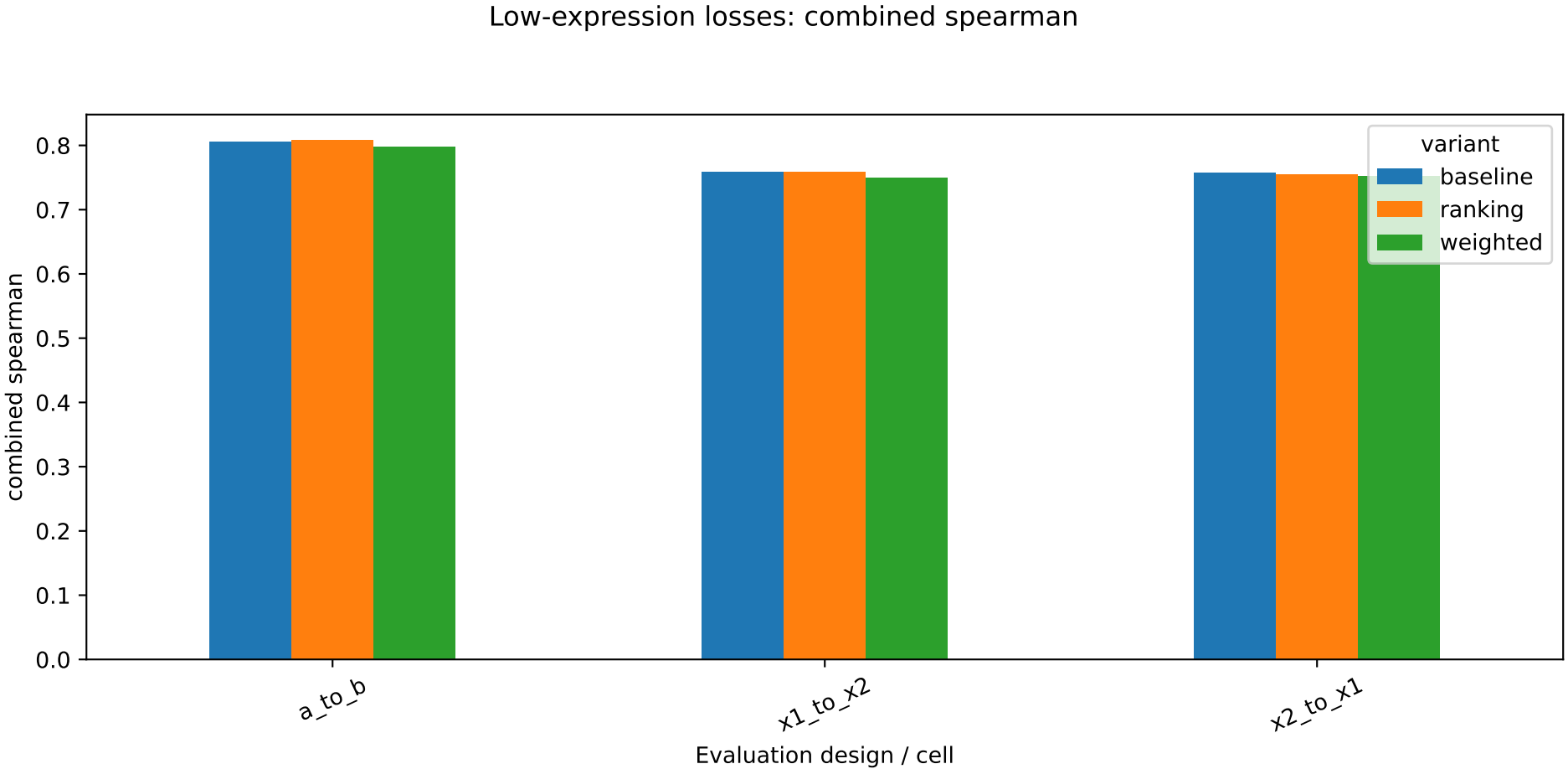
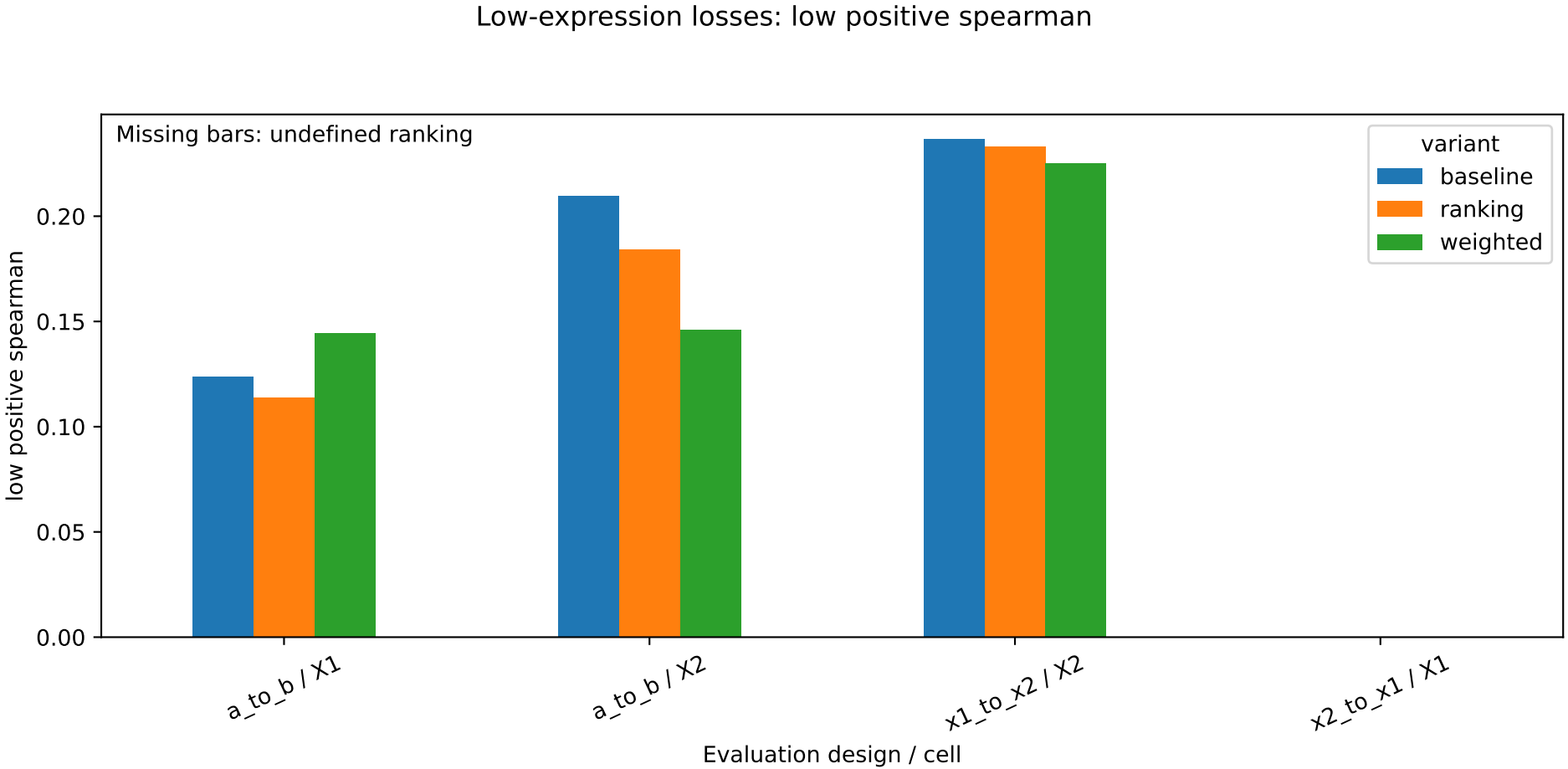
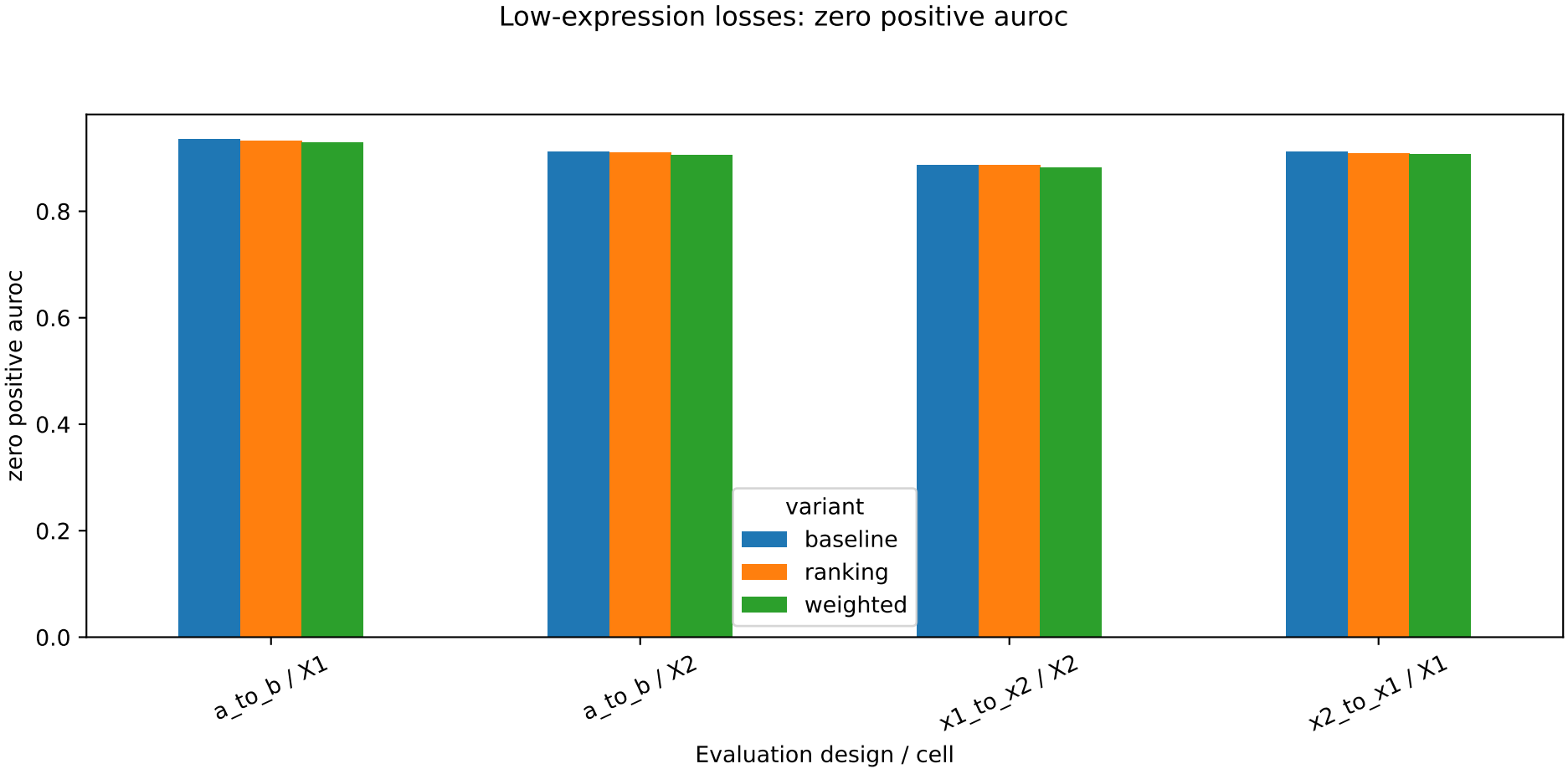
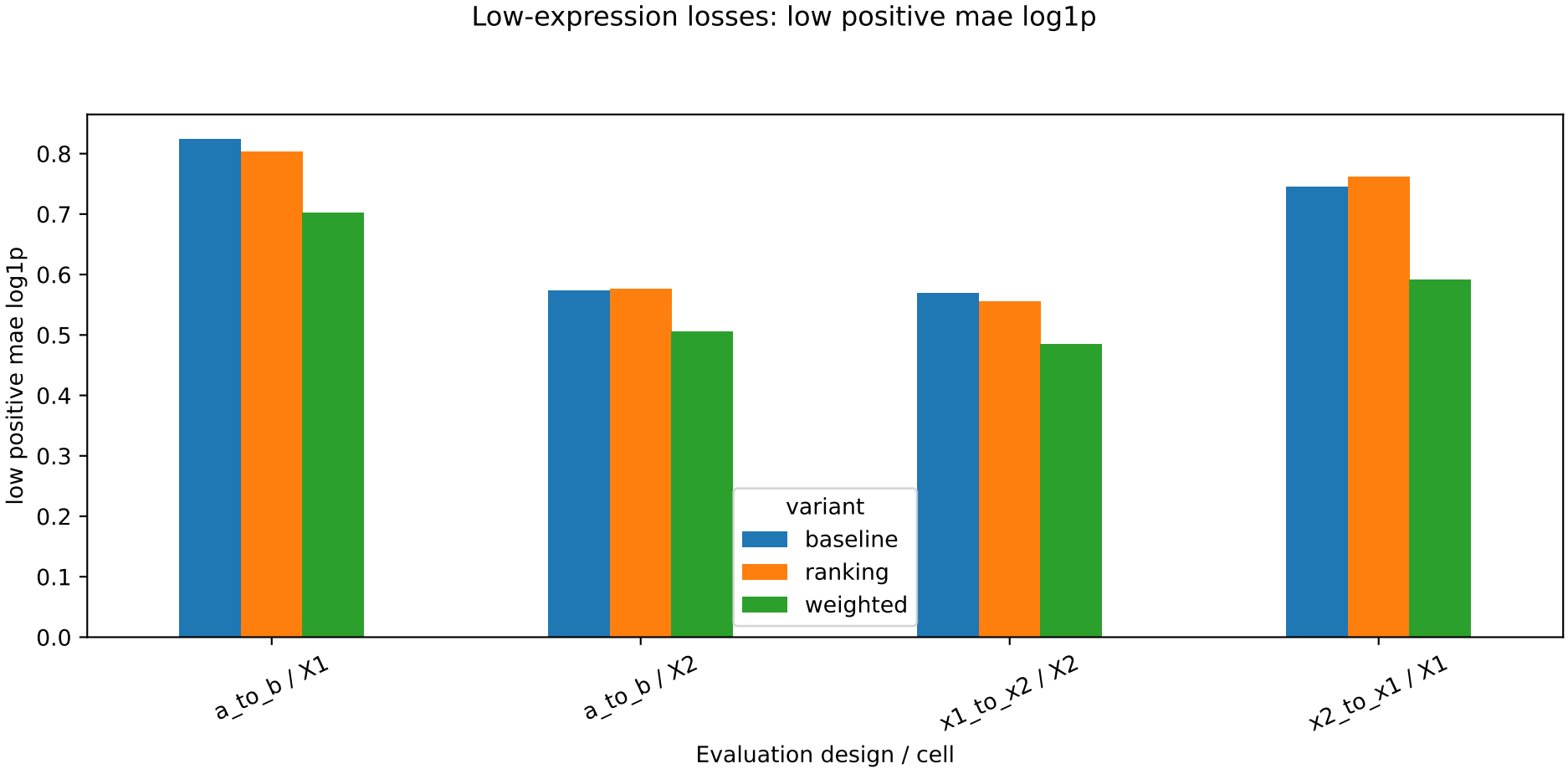
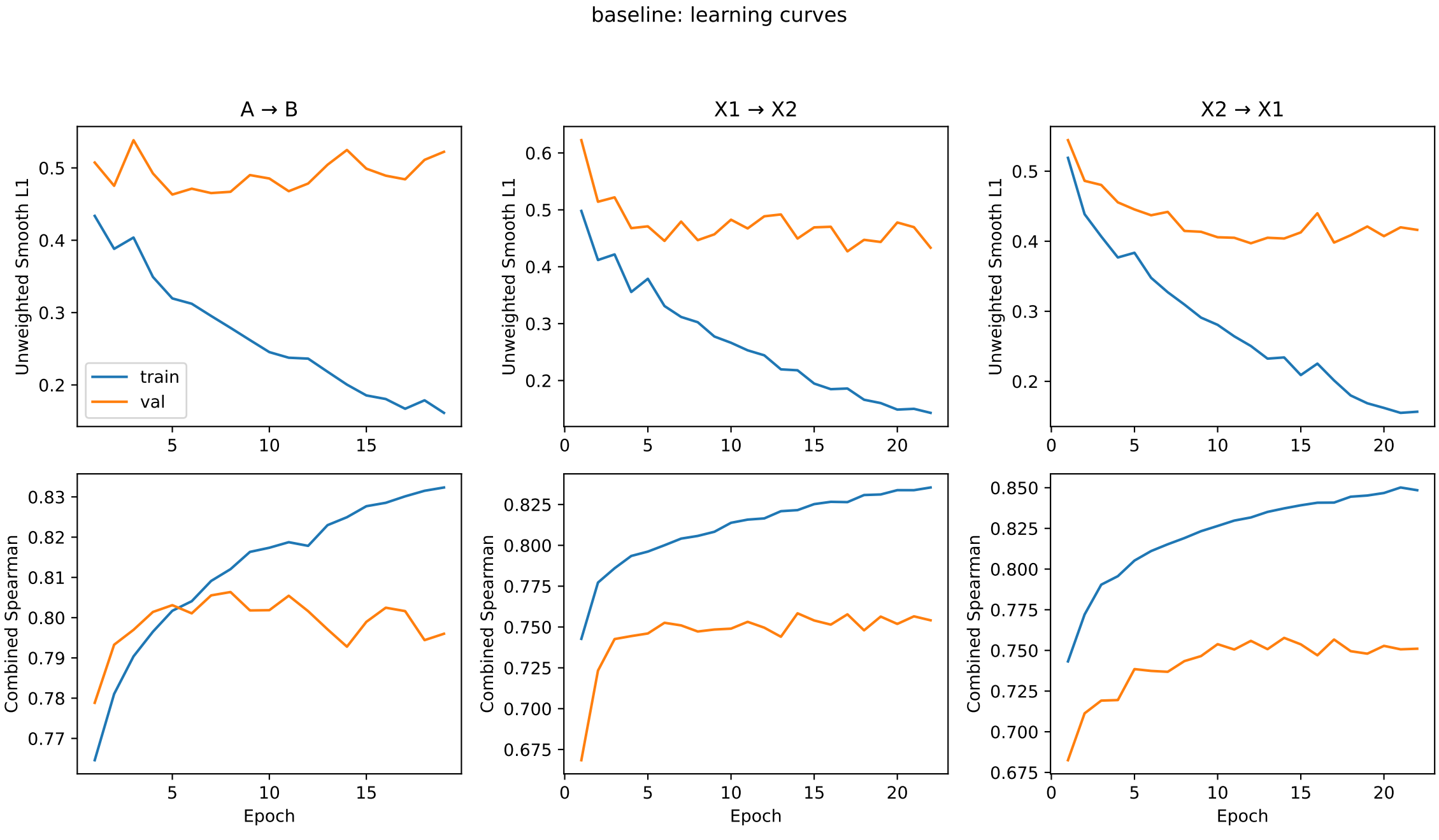
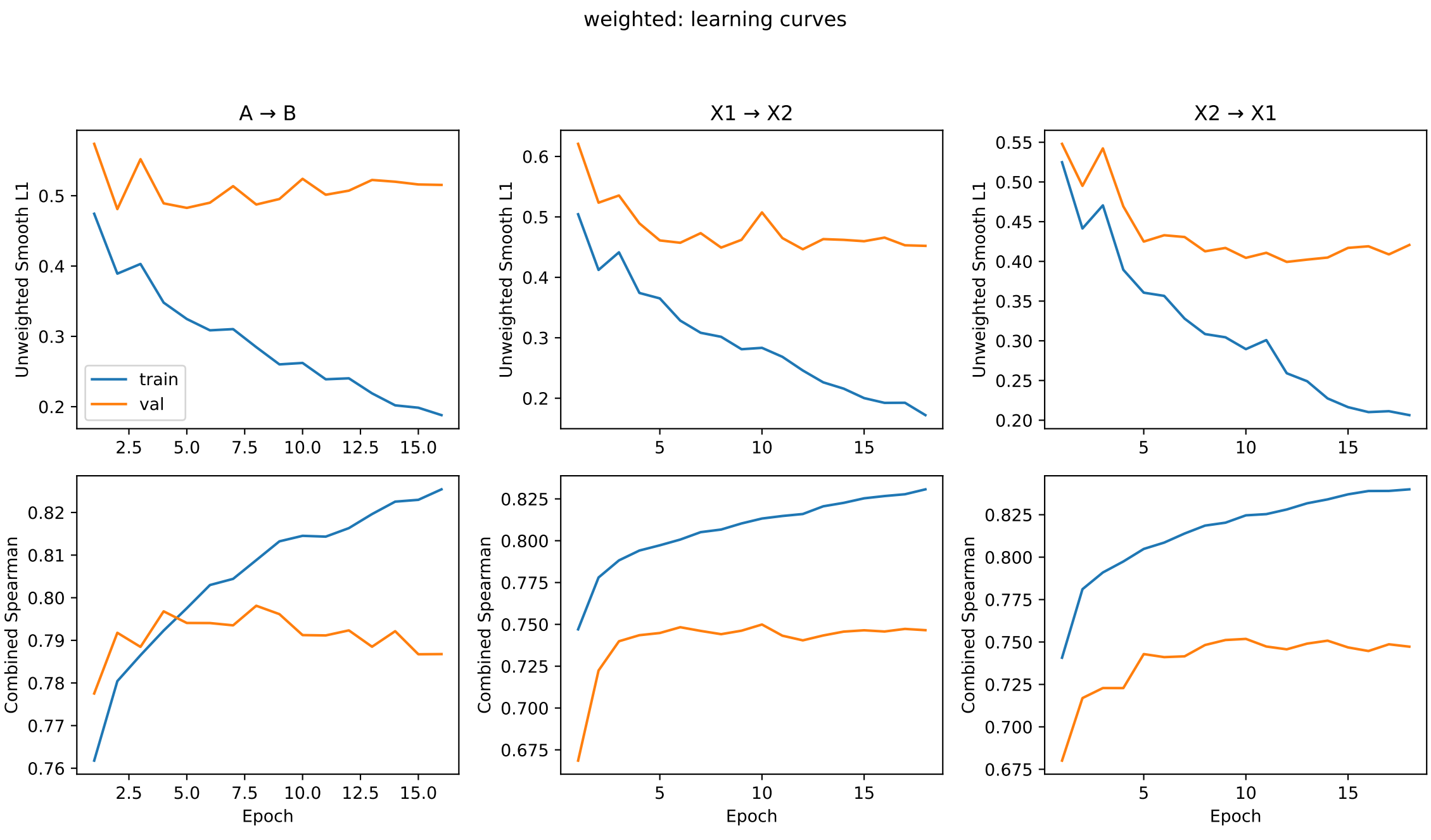

Low-expression losses: combined spearman
combined spearman
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
variant
baseline
ranking
weighted
a_to_b
x1_to_x2
x2_to_x1
Evaluation design / cell


Low-expression losses: low positive spearman
Missing bars: undefined ranking
variant
baseline
ranking
weighted
low positive spearman
0.00
0.05
0.10
0.15
0.20
a_to_b / X1
a_to_b / X2
x1_to_x2 / X2
x2_to_x1 / X1
Evaluation design / cell

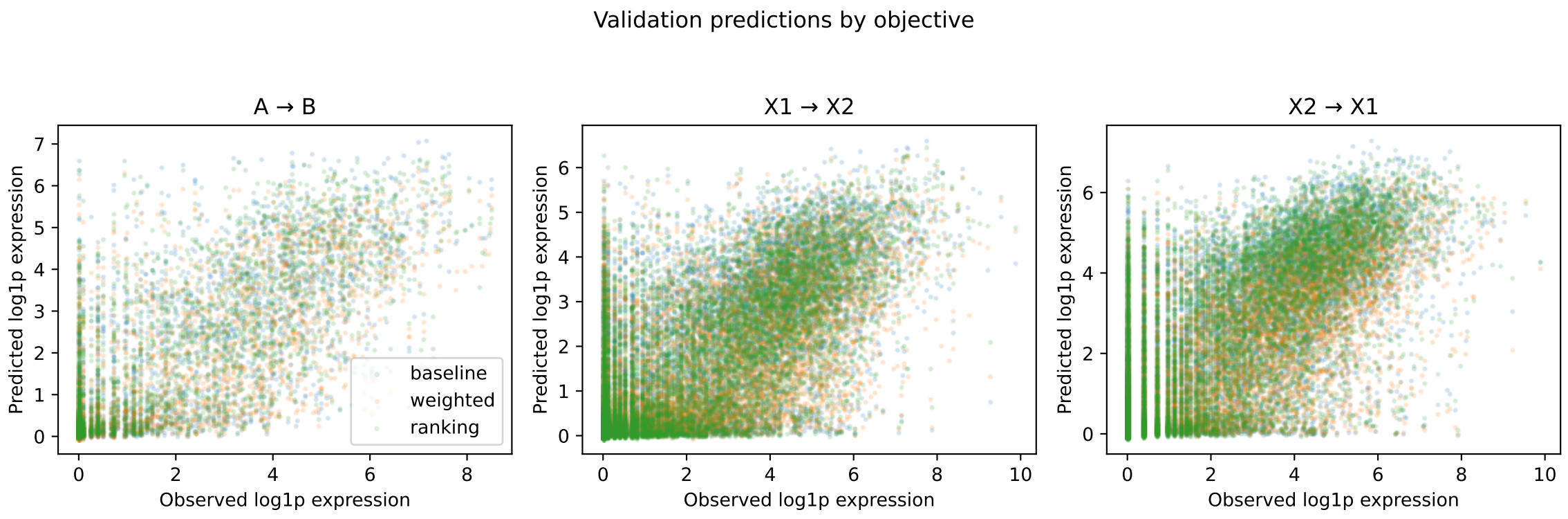

Low-expression losses: zero positive auroc
zero positive auroc
0.0
0.2
0.4
0.6
0.8
variant
baseline
ranking
weighted
a_to_b / X1
a_to_b / X2
x1_to_x2 / X2
x2_to_x1 / X1
Evaluation design / cell


Low-expression losses: low positive mae log1p
low positive mae log1p
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
variant
baseline
ranking
weighted
a_to_b / X1
a_to_b / X2
x1_to_x2 / X2
x2_to_x1 / X1
Evaluation design / cell


baseline: learning curves
A → B
X1 → X2
X2 → X1
Unweighted Smooth L1
Combined Spearman
Epoch
train
val


weighted: learning curves
A → B
X1 → X2
X2 → X1
Unweighted Smooth L1
Combined Spearman
Epoch
train
val


ranking: learning curves
A → B
X1 → X2
X2 → X1
Unweighted Smooth L1
Combined Spearman
Epoch
train
val


Validation predictions by objective
A → B
X1 → X2
X2 → X1
Predicted log1p expression
Observed log1p expression
baseline
weighted
ranking